

Wind Farm	AIC Rank	Distribution	Shape (ξ / k)	Location (μ)	Scale (σ / c)	Log-Likelihood	AIC	BIC	KS Statistic (	KS p-value	Chi ² Statistic
De Aar	1	Gumbel	—	8.2771	0.6692	-78.254	160.509	164.888	0.0635	0.9373	10.313
	2	Frechet	12.6493	—	8.2494	-78.398	160.796	165.176	0.0513	0.9915	9.592
	3	GEV-3P	-0.0166	8.2711	0.665	-78.243	162.486	169.055	0.06	0.9601	10.117
	4	Weibull	9.9261	—	9.0736	-89.611	183.222	187.601	0.1376	0.1496	24.483
Gouda	1	GEV-3P	-0.3213	7.5681	0.7194	-94.405	194.809	194.809	0.0837	0.712	5.741
	2	Frechet	9.5908	—	7.6566	-95.979	195.957	200.337	0.0994	0.5011	8.874
	3	Gumbel	—	7.7053	0.8601	-98.38	200.76	205.14	0.1017	0.4715	11.454
	4	Weibull	5.5691	—	8.8478	-120.815	245.63	250.009	0.2029	0.0074	34.798
Karusa	1	Frechet	11.0723	—	8.8856	-94.922	193.844	198.223	0.1194	0.2806	11.699
	2	GEV-3P	-0.1803	8.8464	0.7789	-94.21	194.42	200.989	0.1118	0.3545	13.643
	3	Gumbel	—	8.9306	0.8505	-97.827	199.654	204.033	0.1371	0.1525	10.763
	4	Weibull	5.1288	—	10.1539	-132.58	269.16	273.539	0.2727	0.0001	64.227
Noupoort	1	Frechet	14.8526	—	11.4621	-89.228	182.456	186.835	0.0822	0.7324	6.865
	2	Gumbel	—	11.4909	0.7892	-89.494	182.987	187.367	0.0714	0.8654	7.053
	3	GEV-3P	-0.0578	11.4661	0.7737	-89.221	184.443	191.012	0.0806	0.7541	6.868
	4	Weibull	8.817	—	12.4924	-113.517	231.034	235.414	0.179	0.0254	37.634
Nxuba	1	GEV-3P	0.1733	13.8998	1.0162	-98.64	209.849	209.849	0.0955	0.5521	9.213
	2	Gumbel	—	13.8067	0.9795	-100.324	204.647	209.026	0.1249	0.2343	13.213
	3	Frechet	14.131	—	13.7702	-101.657	207.314	211.693	0.1331	0.1759	15.483
	4	Weibull	12.9929	—	14.8544	-105.208	214.417	218.796	0.1083	0.3926	12.44
Oyster Bay	1	Frechet	13.9678	—	12.5176	-98.399	200.798	205.177	0.0627	0.9434	1.346
	2	Gumbel	—	12.5534	0.9126	-98.537	201.074	205.453	0.0527	0.9884	1.015
	3	GEV-3P	-0.0433	12.5315	0.901	-98.325	202.65	209.219	0.0585	0.9678	1.157
	4	Weibull	7.7397	—	13.7136	-126.496	256.992	261.372	0.1956	0.011	32.629
Roggeveld	1	Frechet	11.0722	—	8.9915	-95.704	195.409	199.788	0.1194	0.2806	11.699
	2	GEV-3P	-0.1803	8.9519	0.7882	-94.992	195.984	202.553	0.1118	0.3545	13.643
	3	Gumbel	—	9.0371	0.8607	-98.609	201.218	205.598	0.1371	0.1525	10.763
	4	Weibull	5.1288	—	10.275	-133.362	270.725	275.104	0.2727	0.0001	64.227
	1	Frechet	13.0955	—	10.9337	-96.397	196.793	201.172	0.0735	0.8419	2.668

Soetwater	2	GEV-3P	-0.1363	10.9066	0.8218	-96.029	198.059	204.627	0.0718	0.8612	3.363
	3	Gumbel	—	10.9718	0.8687	-98.298	200.595	204.975	0.0825	0.7289	2.391
	4	Weibull	6.3083	—	12.2062	-132.206	268.412	272.792	0.2256	0.002	48.987

Chi ²	p-value	Best model
0.1715		Gumbel
0.2129		
0.182		
0.0009		
0.5703		GEV-3P
0.2618		
0.12		
0		
0.1109		Frechet
0.0579		
0.1493		
0		
0.4431		Frechet
0.4234		
0.4427		
0		
0.2377		GEV-3P
0.0671		
0.0303		
0.087		
0.9872		Frechet
0.9946		
0.9919		
0		
0.1109		Frechet
0.0579		
0.1493		
0		
0.9139		Frechet

0.8495

0.9351

0